# Technical Document #1032: Retrieval Augmented Generation - Comprehensive Overv

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Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

## Key Takeaways:

- Context window optimization selects the most relevant information to fit within the LLM's context limit.
- Chunking splits documents into smaller segments for embedding.
- Hybrid search combines keyword-based and semantic search.